

Agnitia 36h: 9-Hour Plan to the 3 pm Mentor Session

Clock: 6:00 am to 3:00 pm **Team:** you (coordination and backend), Nandani (SUMO), Prakhyat (ML and metrics), Urvashi (dashboard and phone UI) **Based on:** the Connected-Phone Build Plan from Astra, trimmed to what is realistic in 9 hours.

What the mentor should see at 3 pm

A realistic target, not the full product:

1. A SUMO market corridor where a rain slowdown or blocked lane really jams the baseline.
2. Two or three real phones connected to the laptop, each showing "You are [role], a simulated vehicle."
3. A dashboard showing a gray road with no sensor. When phones switch on sharing, probe markers appear.
4. Detection firing from phone data. If time allows, one bounded signal response with a before/after number.
5. The story and the architecture diagram.

ML boosting, CO2 numbers and the full experiment come after the mentor session. Say clearly what is built and what is planned.

Team-wide checkpoints

Time	Gate
7:00	Repo created, contracts frozen, everyone has the example messages
9:00	Nandani's network is valid. Two phones connect to the fake server. Urvashi's phone screen works on fixtures
12:00	H6 gate: the no-incident run clears, the incident run jams

Time	Gate
1:30	A real SUMO vehicle frame reaches a real phone
2:15	Feature freeze. Only fixes and rehearsal until 3:00

You: coordination and backend

Time	Task	Done when
6:00-7:00	Create the repo. Copy the contracts and example messages from Astra's appendices into <code>contracts/</code> . Set up the folder layout and ownership. Share with the team	Everyone can pull and read the fixtures
7:00-9:00	FastAPI server serving fake <code>vehicle.frame</code> messages. Join codes or QR, <code>/api/claim</code> , WebSocket. Test on a hotspot	Two real phones show different roles
9:00-11:00	Single TraCI worker on Nandani's network, with pause and speed control. Stream real vehicle frames	A real SUMO vehicle appears on a phone
11:00-12:30	Probe gateway: phone uplink in, observation out. No uplink means no observation	Turning sharing off removes the observation
12:30-2:15	Simple bounded response (extend green, one advisory to the phone), logged	One response visible in the audit log. Cut it if behind
2:15-3:00	Run the full demo path twice. Record a backup video	Video saved locally

Nandani: SUMO

Time	Task	Done when
6:00-7:30	Install SUMO. Official Hello World tutorial. Pair with Prakhyat	One vehicle completes its route. Version recorded
7:30-9:00	Four-junction diamond with main route and bypass. Write down the actual edge, lane and signal IDs	Both routes complete. ID list shared with the team
9:00-10:30	Car, bike and auto types. Demand. Three role vehicles (<code>veh.role.rider</code> , <code>veh.role.auto</code> , <code>veh.role.delivery</code>). 30-minute sublane trial	Traffic flows. Keep or cut sublane decided
10:30-12:30	Incident: stopped delivery vehicle, or rain speed ramp. Run without and with the incident	H6 gate
12:30-2:00	Detectors. Headless run that writes trip output. Run the fixed baseline on 2-3 seeds	Files Prakhyat can load
2:00-3:00	Screen-record the baseline jam. Write down any bugs	Recording saved

Hard deadline: if the diamond isn't valid by 9:00, switch to the simplest corridor of three or four edges. The jam matters more than the layout.

Prakhyat: ML and metrics

Time	Task	Done when
6:00-7:30	Pair with Nandani on SUMO install and Hello World	Both can run SUMO
7:30-9:30	Journey-time and cohort accounting on a mock CSV, with hand-calculated tests	Tests pass, including one unfinished trip

Time	Task	Done when
9:30-11:30	Observation aggregation and slowdown score. Unknown stays unknown. One vehicle sending many samples counts once	Tests pass on synthetic data
11:30-1:00	Detector plus the simple trend and persistence forecast	Alert fires on a fake jam and not on a normal red light
1:00-2:30	Run it on Nandani's real run logs. Plot slowdown against time. Start the data-generation script	First real plot
2:30-3:00	One slide: what we measure and the preliminary numbers, labelled preliminary	Slide ready

Urvashi: dashboard and phone UI

Time	Task	Done when
6:00-7:00	Storyboard three screens: operator dashboard, phone, live-versus-baseline	Sketch shared
7:00-9:00	Phone UI on fixtures: role banner, sharing toggle, report button, guidance card with Accept and Ignore. "Applying..." until the server confirms	Works on a real phone
9:00-11:30	Dashboard map in SVG: gray unknown road, probe markers, "N phones reporting" counter, permanent labels (SYNTHETIC, SIMULATED, LIVE, RECORDED)	Gray turns observed in the fixture replay
11:30-1:30	Live-versus-baseline layout and audit panel. Switch to the real WebSocket when you have it	Real data on screen
1:30-2:30	Pitch visuals: title, problem story, architecture diagram	Slides ready

Time	Task	Done when
2:30-3:00	Check everything on the big screen	Text readable from the back

Use the Nandipur Bazaar name and keep everything offline, with no online fonts or map tiles.

Rules for everyone's agents

- Give each agent one module, the contracts, and one test. Never say "build the backend."
 - Each person edits only their own folders. Only the coordinator changes the contracts.
 - Commit at least every hour, so nobody loses work if an agent breaks something.
 - Take a 15-minute break around midday. This is a 36-hour event.
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If you fall behind, cut in this order

1. The bounded signal response (show detection only)
2. The third phone (use a laptop tab marked "emulator")
3. Sublane
4. CO2 and boosting (post-mentor anyway)
5. Live baseline recording (use the screen capture)

Never cut the jam, the phone connection and the gray-to-observed reveal. That is the story.

What to ask the mentor

1. Is the "phones as probe devices" framing convincing, or does it feel staged?
 2. What will judges care about most: live demo, numbers, or story?
 3. Is the 10-minute plan too heavy to rehearse?
 4. Are there similar solutions we should watch out for?
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Open item

Turn Astra's appendices into ready files (`contracts/` , example messages, repo skeleton) so the 6:00-7:00 coordinator task takes minutes.